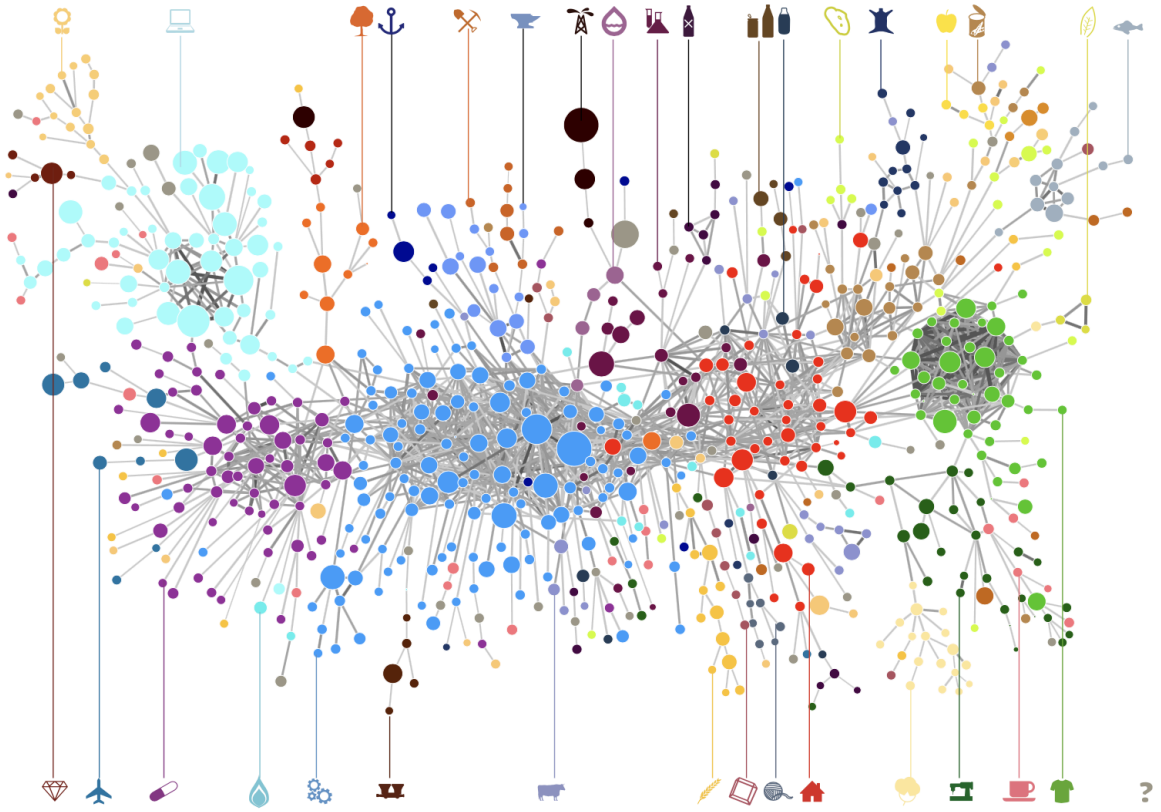


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# Optimal Industry Development Strategy Using Economic and Product Complexity

# United Arab Emirates

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The Economic Complexity Lab - Columbia University

Navia Zhang, Zev McManus-Mendelowitz, Alexandra Garcia, Kelly Mi, Margarita Demkina, Allison Lin, Helena Yang, Jane Zhang,

Daniel Xu, Ryo Shibata, Haoxian Chen, Davepriyan Gunaratnam, Rohan Macherla, Ajit Akole, Shomik Ghose, Alex Wei-Liang Ku, Sannidhya Modi, Xilin Wang

# Optimal Industry Development Strategy for United Arab Emirates

## 0. Executive Summary

In this report, we analyze the economic trajectory of United Arab Emirates over the period from 2019 to 2024, examining its production capability and structural evolution through the lenses of Gross Domestic Product (GDP) and Economic Complexity Index (ECI). We find that United Arab Emirates demonstrated a medium level of economic growth, with an average annual GDP growth rate of 4.9% during this period, placing it in the middle of its emerging peers.

This growth profile reflects a productive structure still anchored by hydrocarbon exports — which continue to account for roughly a third of GDP — alongside an accelerating non-oil economy driven by trade and logistics, financial services, tourism, and construction. Over the analysis window the UAE absorbed a sharp pandemic-era contraction in 2020, benefited from elevated energy prices in 2022, and channelled large state-directed investments into diversification programs tied to the "We the UAE 2031" vision, including expansions of free-zone manufacturing, aviation, and re-export trade through Jebel Ali and Khalifa Port.

On the complexity front, United Arab Emirates's ECI decreased over the analysis period, reflecting a standing below the peer average among the comparable emerging economies examined. The decline in ECI alongside positive GDP growth indicates that the UAE's expansion was concentrated in established re-export and resource-linked categories rather than in the deepening of new productive capabilities, suggesting that aggregate output has outpaced the diversification of underlying export know-how.

Examining realized industry development between 2019 and 2024, we find that United Arab Emirates progressed across the majority of its industry groups: 17 groups moved closer to global competitiveness thresholds while 1 retreated, a increase from 10 improving groups at the start of the period. Based on our opportunity gain and sensitivity analysis, the industries offering the greatest complexity upside for United Arab Emirates are High-complexity Products, Machinery and Construction, and Chemical Manufacturing. Sensitivity analysis further identifies Precious Stones as the sector whose marginal expansion yields the largest positive shift in ECI.

Our Structural Optimality Index (SOI) analysis measures the degree to which the United Arab Emirates's realized investment trajectory aligns with the theoretically optimal complexity-based development path. The United Arab Emirates ranks in the Q4 quartile globally, with an adjusted $R^2$ of 0.0036, indicating a high level of structural alignment. In concrete terms, the United Arab Emirates has lightly channeled resources toward Animals and By-products, Vegetables, Fruits and Plants, Food and Preparations, Energy Drilling and Mining, Wood Products, Textile, Precious Stones, and Metals — sectors where the complexity opportunity is more limited — while underinvesting in Machinery and Construction, Chemical Manufacturing, and High-complexity Products, which offer greater complexity upside. Given the positive correlation between SOI and long-run GDP growth

established across our global sample, the United Arab Emirates's moderate structural alignment is consistent with its medium growth record, and suggests that measurable gains remain available through a more complexity-guided development strategy.

# 1. Report Overview

In this report, we analyze the economic trajectory of the United Arab Emirates during the period from 2019 to 2024. This window captures a pivotal phase for the federation, spanning the launch of the "Projects of the 50" reform package, the post-pandemic recovery, the staging of Expo 2020 Dubai, and the rollout of the We the UAE 2031 strategy, all of which were designed to accelerate the structural pivot from oil revenues toward a more diversified, knowledge- and services-based economy. Specifically, we examine the United Arab Emirates's macroeconomic trajectory through the lens of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect a country's production capability and productive structure, respectively. We then analyze the industrial framework and structural changes of the United Arab Emirates based on quantitative measures, including the Product Complexity Index (PCI), followed by a detailed analysis of how those changes aligned with optimal development strategies as measured by opportunity gains at the industry level. We conclude with macro-level results derived from regression models established on a global scale, which take into account countries' positions in the product space.

## 2. Macroeconomic Results

### 2.1 GDP Ranking and Changes

Figure 1 illustrates the GDP time series for United Arab Emirates and its peer economies (Saudi Arabia, Kuwait, and Qatar) from 2019 to 2024. United Arab Emirates demonstrated a medium average annual GDP growth rate of 4.9% over this period, placing it in the middle of peers among the compared economies. For reference, the average annual growth rates of the peer economies were: Saudi Arabia at 6.8%, Kuwait at 3.9%, and Qatar at 5.1%.

The trajectories diverge most visibly in 2020, when all four economies contracted in absolute terms before resuming growth from 2021 onward; the UAE's recovery was steeper than Kuwait's but slightly flatter than Saudi Arabia's, reflecting the latter's faster rebound in hydrocarbon receipts. By 2024 the UAE's nominal GDP had moved decisively above Kuwait's and Qatar's and held the second-largest absolute level in the peer group, with the gap to Saudi Arabia widening rather than closing as Riyadh's higher growth rate compounded over the period.

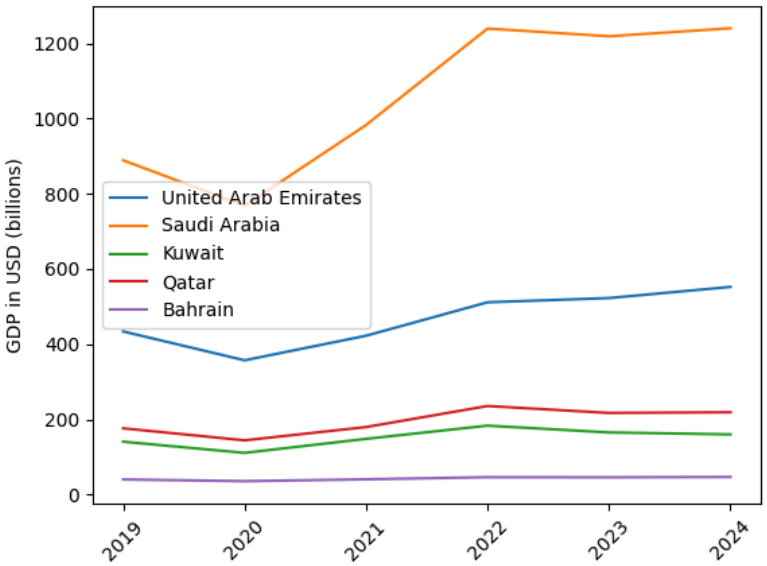



Figure 1. GDP Comparison Among Peer Countries (2019-2024)

### 2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), introduced by Hausmann and Hidalgo (2009) to quantify the knowledge embedded in a country's productive structure, is illustrated in Figure 2.

United Arab Emirates's ECI decreased from 0.675 in 2019 to -0.255 in 2024, a change of -0.930 — a significant shift in productive complexity. Relative to its emerging peers, United Arab Emirates's ECI trajectory was below the peer average. For context, ECI changes over the same period were: Saudi Arabia at +0.456, Kuwait at -0.475, and Qatar at -0.475.

The bulk of the decline is attributable to a recomposition of UAE exports toward lower-complexity categories — notably Precious Stones (gold and refined pearls re-exports), Energy Drilling and Mining, and Metals — which expanded their share of total exports faster than higher-complexity Machinery and Construction and Chemical Manufacturing goods. The ECI movement therefore reflects increased export concentration in a handful of resource-linked and re-export categories rather than a broad-based loss of capability, but it does indicate that the marginal export dollar over 2019–2024 embodied less productive knowledge than at the start of the window.

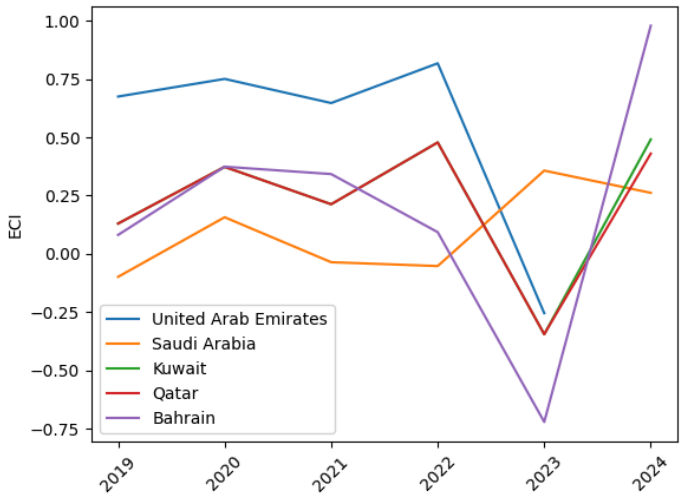



Figure 2. ECI Comparison Among Peer Countries (2019-2024)

Figure 3 provides a combined view of GDP and ECI for the United Arab Emirates across the full analysis window. The two indicators were partially consistent during this period. While GDP rose steadily after the 2020 trough, ECI moved in the opposite direction across most of the same years — diverging most sharply between 2021 and 2023 — implying that the UAE's output expansion was driven primarily by volume in existing export lines rather than by entry into new, more complex product categories, and that the structural depth of the economy did not keep pace with its headline growth.

The 2020 contraction associated with the COVID-19 pandemic represented a material external shock to the UAE's GDP and trade structure: oil prices collapsed, global aviation and tourism volumes (a key non-oil pillar through Emirates and Etihad and through Dubai's hospitality sector) fell sharply, and re-export flows were temporarily disrupted. The ECI series partially absorbed this shock — registering its sharpest single-year drop between 2020 and 2021 — as the rebound favored hydrocarbon and precious-stone re-export categories over the more diversified non-oil manufacturing base.

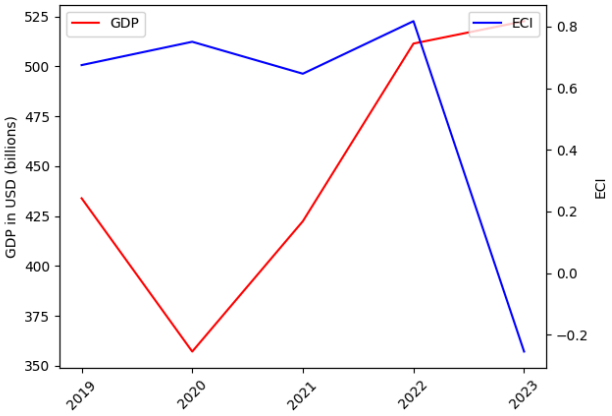



Figure 3. GDP Growth and ECI for United Arab Emirates (2019-2024)

# 3. Production Analysis

## 3.1 Product Categorization

Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Products in our data source are labeled from 010000-999999, with the first two digits representing a general category to which the product belongs. Based on the 99 categories extracted from the label, we further group the products into the 18 most widely-accepted industries, as summarized in Table 1.

Note that the High-complexity Products industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technological gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), the High-complexity Products industry has the highest PfCI among all industry groups.

| Category | Description |
|---|---|
| **Animals and By-products** | All kinds of meat, livestock and related by-products |
| **Vegetables, Fruits and Plants** | All kinds of plants, edible vegetables and fruits |
| **Food and Preparations** | Edible preparations, raw cooking materials and appliances |
| **Clothing and Accessories** | All kinds of fabrics, filament, clothing and accessories |
| **Textiles** | All kinds of textile articles and related products |
| **Wood Products** | Book, paper and other elementary products related to wood |
| **Metals** | Mining of all sorts of raw Metals |
| **Refined and Processed Stones** | Refined or processed stone products such as pearl and ceramics |
| **Energy Drilling and Mining** | Mining industry for energy, minerals, stone materials, etc. |

| | |
|---|---|
| **Daily Instruments** | Basic daily instruments such as sport requisites, timing and measuring tools |
| **Raw Manufacturing** | Basic products made from plastic, rubber, leather, etc. |
| **Chemical Manufacturing** | Advanced chemical products and pharmaceutical products |
| **Chemicals** | Raw chemical materials and by-products |
| **Machinery and Construction** | Machinery and mechanical appliances, including electrical equipment |
| **Transportation** | Vehicles and locomotives including aircraft, ship, railway, etc. |
| **Weapons** | Explosives, arms and ammunition |
| **Services and Utilities** | Financial, business, legal, communication, travel Services and Utilities, etc. |
| **High-complexity Products** | Agglomeration of advanced products with highest PCIs |

Table 1. Categorization of Products

We further observe a positive correlation between the GDP and the export of High-complexity Products. A higher adjusted R-square exists between GDP and High-complexity Products exports, indicating that more successful economics, as measured by GDP, tend to be those that are capable of producing and exporting High-complexity Products as we define them. Tables 2a and 2b show the OLS results in a universe of 107 and 21 typical countries respectively.

| | High-complexity Products | Food and Preparations | Animals and By-products |
|---|---|---|---|
| Adj. R-squared | 0.075 | 0.006 | 0.002 |
| F-statistic | 9.556 | 1.638 | 1.175 |
| No. Observations | 107 | 107 | 107 |

Table 2a. OLS Regression Results of GDP against Percentage Volume of Exports

| | High-complexity Products | Food and Preparations | Animals and By-products |
|---|---|---|---|
| Adj. R-squared | 0.149 | 0.003 | -0.012 |
| F-statistic | 4.510 | 1.059 | 0.7542 |
| No. Observations | 21 | 21 | 21 |

Table 2b. OLS Regression Results of GDP against Volume of Percentage Exports

### 3.2 Realized Industry Development

Based on the aforementioned product categorization, we aim to capture United Arab Emirates's realized industry production strategy between 2019 and 2024 on a complexity level. In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product, not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop the corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is of developing products within that group. Defining this net distance change as the Realized Industry Development (RID), our export data allow us to calculate and compare the RID of United Arab Emirates and the 18 product groups in 2024 and 2022 through the difference, where a positive RID indicates a shortening of the distance (i.e., advancement of the industry), and a negative RID indicates the reverse. It is notable that the number of industries with positive RID experienced a increase from 10 to 17 from 2022 to 2024.

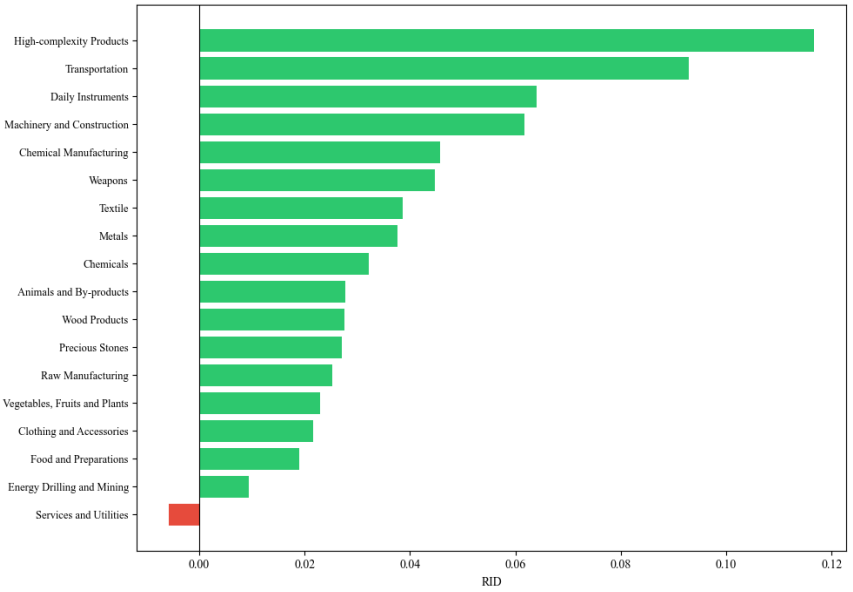



Table 3. Realized Industry Development (RID) of United Arab Emirates

### 3.3 Regional and Global Comparison

Table 4a below summarizes how the production strategy adopted by the United Arab Emirates compares with selected leading economies in the world across different years, as measured in RID. Leading economies such as Germany, Japan, and the Republic of Korea gave considerable attention to more advanced product groups such as Machinery and Construction and High-complexity Products in 2024, and to Chemical Manufacturing, Transportation, and Daily Instruments in 2022.

| Rank | France | Greece | Portugal |
|---|---|---|---|
| #1 | Precious Stones | Chemicals | Precious Stones |
| #2 | Wood Products | Weapons | Chemical Manufacturing |
| #3 | Vegetables, Fruits and Plants | Chemical Manufacturing | Wood Products |
| #4 | Food and Preparations | High-complexity Products | Energy Drilling and Mining |
| #5 | Energy Drilling and Mining | Clothing and Accessories | Food and Preparations |

Table 4a. RID Comparison across Leading Economies (2024)

Table 4b further examines the United Arab Emirates's development strategy against those of emerging economies on a global scale. Reasonably, emerging economies such as Saudi Arabia, Turkey, and Malaysia have devoted the majority of their efforts to intermediate industry groups, showing moderate shifts in strategies over the course of the analysis period.

| Saudi Arabia | Turkey | Malaysia | |
|---|---|---|---|
| nery and Construction | Food and Preparations | Chemical Manufacturing | Chemical M |
| h-complexity Products | Energy Drilling and Mining | High-complexity Products | Machinery and |
| Transportation | Clothing and Accessories | Machinery and Construction | |
| Weapons | Animals and By-products | Energy Drilling and Mining | Daily |
| Raw Manufacturing | Wood Products | Chemicals | |

Table 4b. RID Comparison across Emerging Economies (2024)

# 4. Product Space Design

## 4.1 Suggested Development Vector

As outlined above, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined in Hausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP = 0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter of the product as compared to the global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in terms of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP = 1) has an opportunity gain of zero. An unreasonable investment in a certain product p would lead to a relative decrement in the export of more profitable products and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economic strength, those having higher ECI, or in other words, more reasonable economic structures are often in better shape.

| Arab Emirates | Saudi Arabia | Kuwait | Qatar | |
|---|---|---|---|---|
| plexity Products | High-complexity Products | High-complexity Products | High-complexity Products | High-comple |
| Transportation | Services and Utilities | Transportation | Transportation | Vegetables, Fru |
| ily Instruments | Transportation | Daily Instruments | Daily Instruments | T |
| nd Construction | Machinery and Construction | Services and Utilities | Services and Utilities | Food and |
| Chemicals | Daily Instruments | Machinery and Construction | Machinery and Construction | Service |
| Manufacturing | Weapons | Chemicals | Chemicals | Machinery and |
| l Manufacturing | Energy Drilling and Mining | Raw Manufacturing | Raw Manufacturing | Daily |
| Weapons | Chemicals | Chemical Manufacturing | Chemical Manufacturing | |
| and Accessories | Raw Manufacturing | Weapons | Weapons | Chemical M |
| Metals | Chemical Manufacturing | Clothing and Accessories | Clothing and Accessories | |
| Textile | Clothing and Accessories | Metals | Metals | W |
| lling and Mining | Textile | Energy Drilling and Mining | Energy Drilling and Mining | Clothing an |
| Precious Stones | Metals | Textile | Textile | Raw M |
| es and Utilities | Wood Products | Wood Products | Wood Products | Animals and |
| Wood Products | Precious Stones | Precious Stones | Precious Stones | Energy Drillin |
| nd By-products | Animals and By-products | Animals and By-products | Animals and By-products | Pr |
| ruits and Plants | Vegetables, Fruits and Plants | Food and Preparations | Food and Preparations | |
| nd Preparations | Food and Preparations | Vegetables, Fruits and Plants | Vegetables, Fruits and Plants | |

Table 5. Opportunity Gain Ranking on a Comparable Scale (2024)

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 summarizes how the SDV of United Arab Emirates compares with a selected group of comparable and leading economies in the world, with the industry groups ranked in decreasing order of significance.

Dark blue indicates positive opportunity gains, light blue indicates zero opportunity gains, and white indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain and share common interests in High-complexity Products and Transportation industries. It can be concluded that the United Arab Emirates shares the characteristics of developing countries, with a balanced economic structure that leans towards intermediate-level industries.

| Saudi Arabia | Kuwait | Qatar | Oman |
|---|---|---|---|
| High-complexity Products | High-complexity Products | High-complexity Products | High-complexity Products |
| Services and Utilities | Transportation | Transportation | Transportation |
| Transportation | Daily Instruments | Daily Instruments | Services and Utilities |
| Machinery and Construction | Services and Utilities | Services and Utilities | Machinery and Construction |
| Daily Instruments | Machinery and Construction | Machinery and Construction | Vegetables, Fruits and Plants |
| Weapons | Chemicals | Chemicals | Chemicals |
| Energy Drilling and Mining | Raw Manufacturing | Raw Manufacturing | Daily Instruments |
| Chemicals | Chemical Manufacturing | Chemical Manufacturing | Food and Preparations |
| Raw Manufacturing | Weapons | Weapons | Chemical Manufacturing |
| Chemical Manufacturing | Clothing and Accessories | Clothing and Accessories | Weapons |
| Clothing and Accessories | Metals | Metals | Precious Stones |
| Textile | Energy Drilling and Mining | Energy Drilling and Mining | Clothing and Accessories |
| Metals | Textile | Textile | Wood Products |
| Wood Products | Wood Products | Wood Products | Textile |
| Precious Stones | Precious Stones | Precious Stones | Animals and By-products |
| Animals and By-products | Animals and By-products | Animals and By-products | Energy Drilling and Mining |
| Vegetables, Fruits and Plants | Food and Preparations | Food and Preparations | Raw Manufacturing |
| Food and Preparations | Vegetables, Fruits and Plants | Vegetables, Fruits and Plants | Metals |

Table 6. SDV of Other Sample Countries (2024)

### 4.2 Budgeted Industry Development Strategy and Interpretation

The optimal budgeted industry development strategy aims to solve the most efficient allocations under the budgeted constraint of different situations of heuristic total export growth. In order to obtain such an allocation portfolio, we formulate an optimization problem as the following:

$$\underset{x \in \mathbf{R}^{|P|}}{\text{Max}} \quad \left(\frac{k(x) - < k(x) >}{stdev(k(x))}\right)_C$$

$$\text{Subject to}$$

$$\sum_{p \in P} x_p \leq B$$

$$x \succcurlyeq 0$$

where k(x) is the second largest eigenvector of $M_{CC'}$ after adding $x$ to the original export vector, $C$ is the country that we are interested in, $P$ is the product space, $B$ is the heuristic budget.

If we can solve this problem, then we can know what the largest ECI to which a country can improve. Accordingly, the optimal solution $x$ * that we obtain here is exactly our optimal allocation. However, the biggest problem is that the objective function is not convex. Therefore, we cannot apply the classical tools for convex optimization to this problem. In order to solve this problem, our idea is to use a greedy algorithm. The way that we apply the greedy algorithm is that firstly we separate the total budget into k equal parts, and then for the

first equal part we try to search for the product that can optimize our objective function by iterating through each $p \in P$. After we find the optimal product p for this equal part, we add the amount back to the export value of this product p and update the ECI for this country. If we observe that ECI does not change after iterating all the products, then we can add up one more equal part to invest. Until we find there is an ECI increment, we add the amount now to that product. The equal rest parts can be done in the same way.

By definition, a greedy algorithm is a kind of local search algorithm that is possible to run into a local minimum. Therefore, for local search algorithms, one typical way to keep it away from the local minimum is to enhance the vision of the algorithm by looking more steps forward. In our setting, whenever we obtain an increment of ECI, we mark down the required amount of export growth. Simultaneously, we continue searching for the next ECI increment point, compare the ratios of these two possibilities and pick the largest one. The ratio is defined as the following:

$$Ratio(p) = \frac{ECI_p - ECI_0}{||x_p - x_0||_1}$$

where $ECI_p$ is just the ECI of the country after adding some amount of export increment to product $p$, which is exactly the amount that is able to make ECI change. Also, we can express it in the way of $||x_p - x_0||_1$.

In this way, we can obtain a further sighted greedy algorithm. Also, the number of parts that we would like to separate the budget into also matters. In order to solve this problem, we added a stabilizer on top of the enhanced greedy algorithm by running the algorithm for different numbers of parts and selecting the portfolio with the largest ECI increment and most complex economic structure.

Besides, it is quite common that there is a surplus in the budget. To deal with the surplus, we just add it to the product which needs the least amount of dollars to make $M_{cp} = 1$ (i.e., the smallest amount of required export increase).

### 4.3 Economic Panel

As stated above, the enhanced greedy algorithm is developed mainly for solving the optimal allocation strategy. In order to visualize this allocation, an economic panel is created, consisting of a pie chart, a line chart and a table. First, a pie chart is used to show the percentage of each product in the optimal portfolio. The number located in the center of the circle denotes the heuristic growth of total export, which is also our budget. Each category in the chart has a determined percentage of the allocated budget. The lines below the pie charts show how the ECI changes during the searching process of the greedy algorithm. Even though sometimes an increase of the export by 15% or 20% does not seem reasonable to developed countries, yet it can also give a sense of how a country should develop in the long term. Eventually, the graph in the end is just for numerical explanation for the pie charts.

### 4.5 Interpretation

For the purpose of a clear explanation of the line charts, we label all the products in Table 7. Please refer to this table when trying to understand the investment trajectories in the line charts.

| Label | Product Names |
|---|---|
| 1 | Animal and By-products |
| 2 | Vegetables, Fruits and Plants |
| 3 | Food and Preparations |
| 4 | Energy Drilling and Mining |
| 5 | Chemicals |
| 6 | Chemical Manufacturing |
| 7 | Raw Manufacturing |
| 8 | Wood Products |
| 9 | Clothing and Accessories |
| 10 | Textile |
| 11 | Metals |
| 12 | Machinery and Construction |
| 13 | Transportation |
| 14 | Weapons |
| 15 | Daily Instruments |
| 16 | Precious Stones |
| 17 | High-complexity Products |
| 18 | Services and Utilities |

Table 7. Products Labeling

Figure 4 shows an economic panel for the United Arab Emirates in 2024. ECI gains were not evenly distributed across budget thresholds: the largest marginal improvement in complexity occurred between the 5% and 10% export-growth scenarios, after which incremental returns flattened markedly, indicating a step-change at moderate budget levels rather than a linear payoff. This pattern suggests that a meaningful but achievable expansion in targeted high-complexity exports could capture most of the available complexity upside, while pushing toward 15–20% budget increases yields diminishing structural gains.

For 2024, a 5 percent increase in exports would emphasize High-complexity Products, followed by High-complexity Products again as the dominant allocation at the 10, 15, and 20 percent export-growth scenarios. These strategies are in line with the suggested trajectories based on ECI. The persistence of High-complexity Products as the priority allocation across all budget levels — rather than a rotation toward secondary industries as the budget grows — indicates that the UAE's complexity bottleneck is heavily concentrated in this single high-PCI cluster, and that secondary industries such as Machinery and Construction and Chemical Manufacturing only begin to receive material allocation once the High-complexity Products threshold for additional MCP = 1 status is saturated.

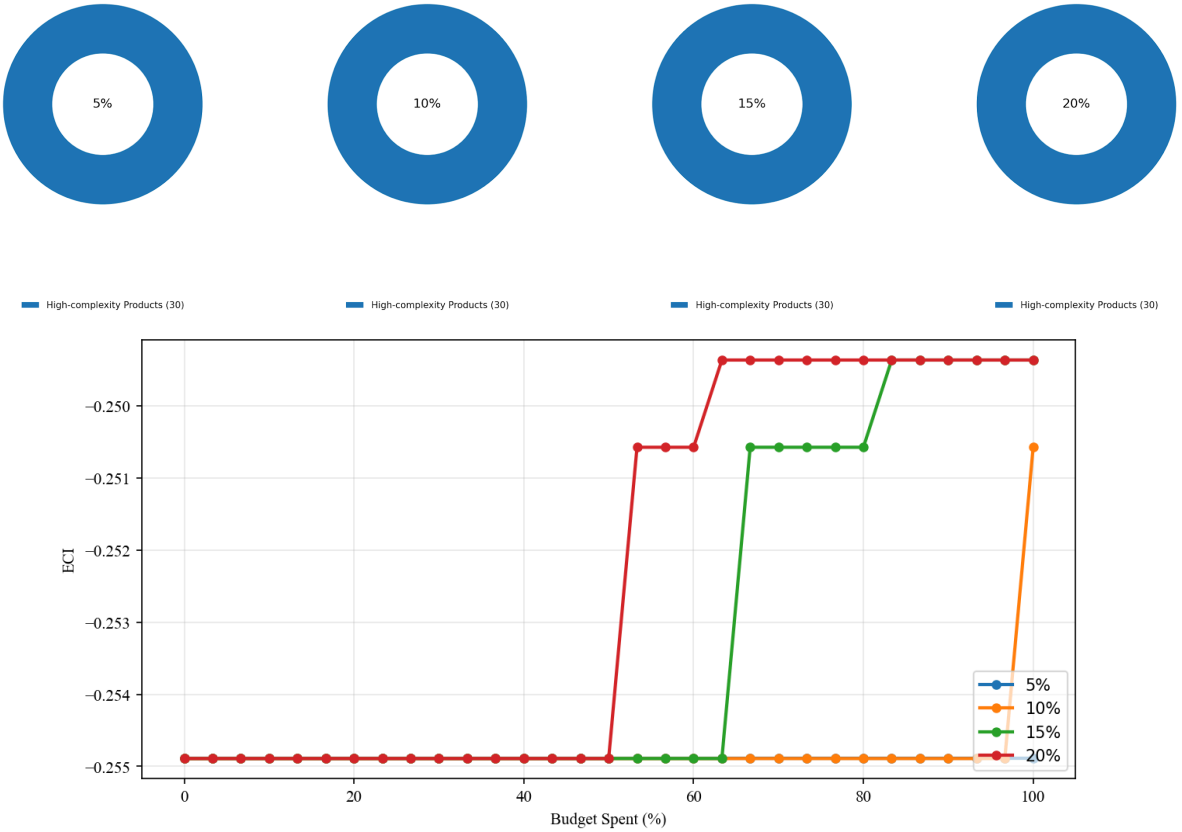



Figure 4. Greedy-Based ECI Improvement Results of United Arab Emirates with Different Budgets (2024)

| labels | 5% | 10% | 15 |
|---|---|---|---|
| Export Growth | $30.5B | $61.0B | $91. |
| lexity Products | $30.49B | $60.97B | $91.4 |
| reased ECI (Δ) | -0.2549 | -0.2506 | -0.24 |

Table 8. Portfolio of United Arab Emirates (2024)

### 4.6 Sensitivity Analysis and Contingency Strategy

In the following section, our broad aim is to understand how sensitive United Arab Emirates is to changes in export in various industries. While the Economic Complexity Index has been shown to contain valuable information about the economic strength/capacity of a country, little has been said as to the stability of the measure itself. We seek to uncover the main drivers of stability under a country's ECI score and better understand longer-term projections

for ECI growth. Additionally, gaining an intuition on the industries that are bolstering an economy and the industries that offer the greatest return per unit of investment is best done through visualization of industry-specific sensitivity. To this end, we have developed two categories of sensitivity analysis.

### 4.7 ECI Sensitivity Graph

First, in order to assess the effect particular industries have on each respective country, we developed ECI Sensitivity graphs (Figure 5). We use the shocking methodology to increment or decrement an industry's export level by a certain percentage and recalculate the ECI, all else being equal. While freezing the export dynamics of the rest of the world and enacting an instantaneous shock in export for a particular industry and country may not reflect real-life dynamics, what it does offer us is an understanding of a country's level of dependency on an industry in the context of ECI. The shocking is done in small percentage increments in order to smooth out the resulting curve. Additionally, a recurring theme one will see is that for most industries, a breaking point occurs where the ECI either jumps upwards or downwards, creating a hockey stick shape rather than a linear trend. While the inflection point may not be accurate in the context of real life, comparing the various inflection points across industries can give us valuable insight.

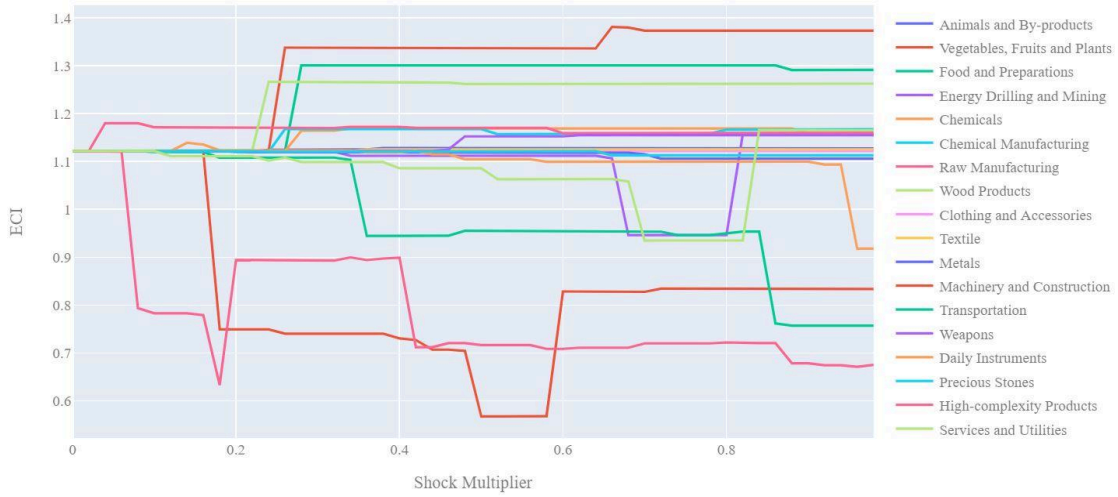



Figure 5. Sample ECI Sensitivity Graph

The graphs are formulated as follows. The first chart, Incremental ECI Sensitivity, shows positive shocking in which export numbers are shocked upwards per industry. The X-axis is in terms of percentage shock per current dollar value of export. The Y-axis is the recalculated ECI level post-shocking. The second chart, Decremental ECI Sensitivity, follows the same format except the X-axis represents negative shock. In other words, exports for each industry are decremented, and the ECI is accordingly recomputed.

### 4.8 Sensitivity Analysis Results and Interpretation

We examine United Arab Emirates's ECI Decremental and Incremental Sensitivity Charts for 2024. An increase in Precious Stones will lead to the highest change in ECI, followed by Metals and Animals and By-products. Under positive shocks, ECI decreases with additional investment in Precious Stones, confirming it as the primary lever for complexity growth. For negative shocks, Energy Drilling and Mining emerges as the most impactful industry, while reducing investment in Precious Stones causes ECI to move upward. The rest of the industries do not appear to have the same level of sensitivity.

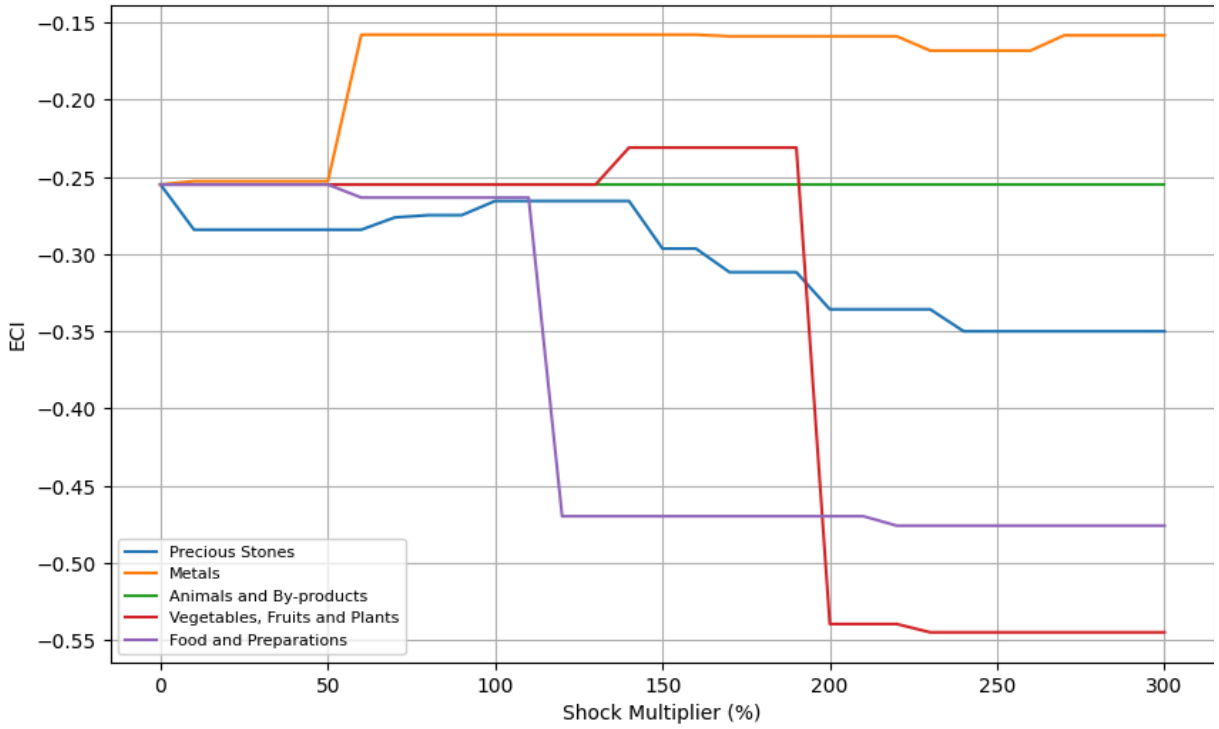



Figure 6a. Incremental ECI Sensitivity for United Arab Emirates (2024)

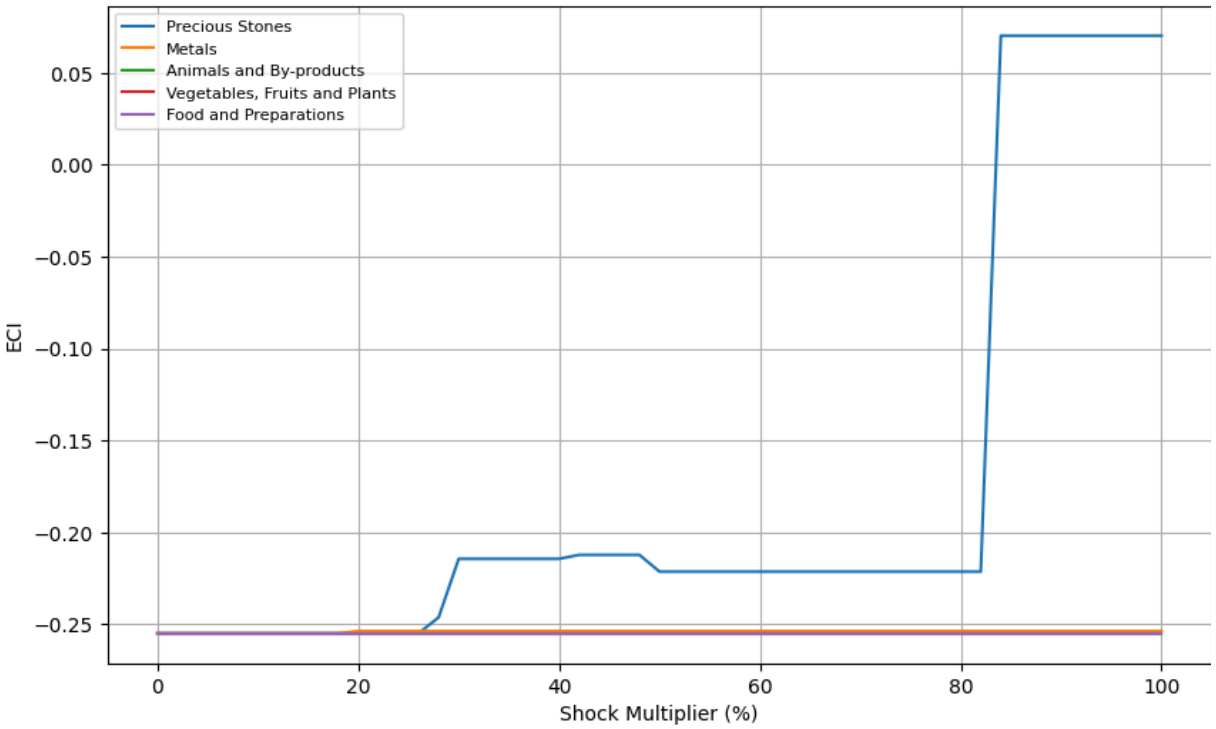



Figure 6b. Decremental ECI Sensitivity for United Arab Emirates (2024)

# 5. Macro-level Predictions

## 5.1 Structural Optimality Index

ECI sensitivity described in the section before has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our development strategies are and how deviations from the suggested developmental path will lead to suboptimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the Adjusted R-Square obtained from regressing the country's realized industry development over the suggested development strategy. Both strategies are quantified in nature, with the realized development represented by increasing export in every major industry group and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The adjusted R-square from the regression thus provides us with a quantified measure of how the development path adopted by the country has fitted with/differed from calculations on a complexity level.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which is defined as the percent increase in GDP (USD), over their SOI. Figure 9 illustrates the regression output, which shows a strong positive correlation between countries' SOI and their five-year GDP growth, achieving an adjusted $R^2$ of 0.0036%. In other words, the more a country develops, as suggested by calculations on a complexity level, the better economic outcomes it achieves.

| **Table 14: OLS Regression Results** | |
|---|---|
| **Dep. Variable:** GDP | **F-Statistic:** 1.4978 |

| | |
|---|---|
| **$R^2$:** 0.0109 | **Prob (F-Stat):** 0.2231 |
| **Adj $R^2$:** 0.0036 | **No. Observations:** 138 |

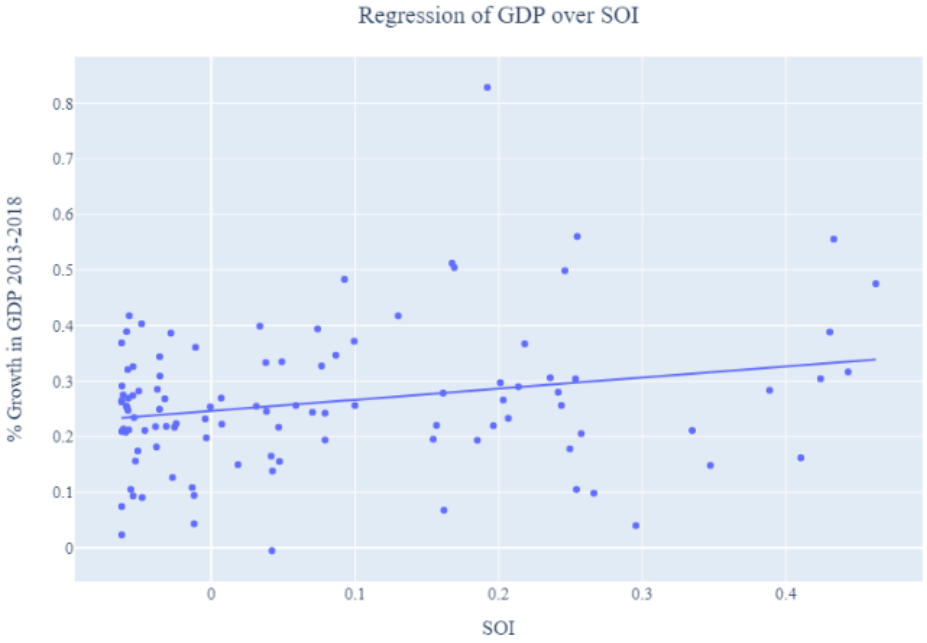



Figure 9. Regression of Five-Year GDP Growth over SOI (2019-2024)

**5.2 SOI of United Arab Emirates and Its Implications**

During the examined period, United Arab Emirates has a high level of structural optimality as compared with all countries for which data is available. As the regression in Figure 10 suggests (the red curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), United Arab Emirates demonstrates a high level of correlation with the suggested developmental path, having an adjusted R-square of around 0.0036.

The United Arab Emirates ranks in the Q4 quartile among all countries in terms of SOI and recorded an average annual GDP growth rate of 4.9% from 2019 to 2024, placing it in the middle of its emerging peers — consistent with the regression result in Figure 9. By contrast, a country such as Kuwait sits in a lower SOI quartile and posted a slower 3.9% annual growth rate over the same period, illustrating the cross-country pattern in which weaker alignment with the complexity-based developmental path is associated with weaker GDP outcomes.

Specifically, United Arab Emirates's medium economic progress from 2019 to 2024 can be at least partially explained by a development strategy that demonstrates a high level of alignment with our suggested developmental path. In particular, United Arab Emirates has lightly channeled resources toward Animals and By-products, Vegetables, Fruits and Plants, Food and Preparations, Energy Drilling and Mining, Wood Products, Textile, Precious Stones, Metals — sectors where the complexity opportunity is more limited — while underinvesting in None, which offer greater complexity upside.

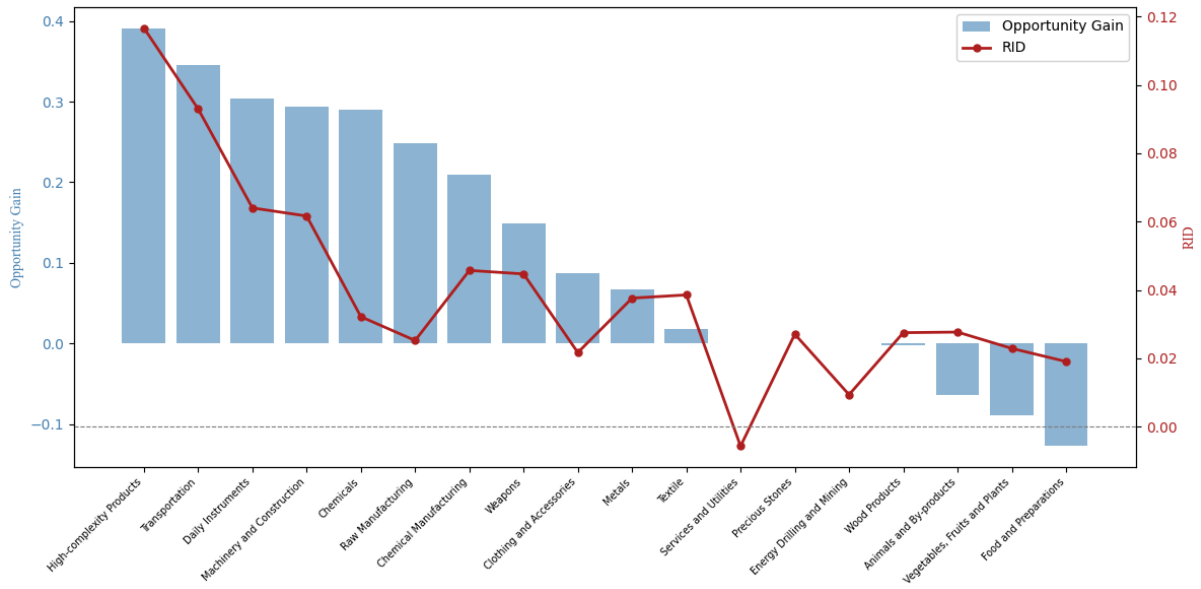



Figure 10. SOI of United Arab Emirates (2019-2024)

## 6. Concluding Remarks

In conclusion, United Arab Emirates has demonstrated a medium level of economic growth in terms of annual GDP increase during the period from 2019 to 2024, concurrent with a significant shift in economic complexity as compared with its peer economies. An average annual GDP growth rate of 4.9% placed United Arab Emirates in the middle of its emerging peers, and ECI decreased by -0.930 over the same window.

We believe that United Arab Emirates's trajectory can be partially explained through the analysis of the economic and product complexity structure on the industry group level. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that United Arab Emirates progressed in developing the majority of its industries during the period, with the count of positively developing industries increasing from 10 to 17 between 2022 and 2024.

Based on our analysis of opportunity gain and sensitivity across major industry groups, United Arab Emirates can further its gains by prioritizing High-complexity Products, , and . Sensitivity analysis further identifies Precious Stones as the sector whose marginal expansion yields the largest positive shift in ECI. We further note that United Arab Emirates has lightly channeled resources toward Animals and By-products, Vegetables, Fruits and Plants, Food and Preparations, Energy Drilling and Mining, Wood Products, Textile, Precious Stones, Metals while underinvesting in None, indicating a high level of structural alignment with complexity-guided development, having an adjusted R-square of 0.0036. Since a positive correlation exists between five-year GDP growth and the structural optimality index as we define and calculate, United Arab Emirates's moderate structural alignment is consistent with its medium growth record, and suggests that measurable gains remain available through a more complexity-guided development strategy.

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